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Course: Data Analytics with Python

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Research Assignment (The Language of Business Analytics)

1. Foundational Concept of Analysis:

• Business analytics: refers to the statistical methods and computing technologies for processing, mining and visualizing data to uncover patterns, relationships and insights that enable better business decision-making.

↳ For example, Netflix uses data analytics to drive content recommendations, where they analyse viewing history and search patterns to predict user preferences.

• Data analytics: is the practice of looking at raw data in various ways to gain information. The broad process of inspecting, cleaning, transforming, and modeling data to discover useful information.

↳ For example, A scientist analyzing hospital patient records to identify the most effective recovery treatments.

• Business Intelligence (BI): a set of strategies and technologies used ~~by~~ for collecting, managing and analysing organisational data to yield insights that inform business strategies and operations. Analysing business information, focusing on what happened.

↳ For example, a clothing retailer analysing sales data, weather, and local events to forecast demand, reducing overstock and preventing lost sales.

• Data-Driven Decision Making: the practice of making organisational decisions based on actual data rather than intuition or observation alone. It involves

using facts and data analysis to guide strategic business actions, replacing intuition with evidence.

↳ For example, Financial institutions use historical data to perform risk assessments, identify fraudulent activity patterns, and create accurate financial forecasts.

- Big Data: refers to large, ~~and~~ complex, and rapidly growing data sets that traditional data processing software cannot manage. ~~An~~ Large dataset analysed to reveal patterns, trends, and associations, particularly relating to human behaviour and interactions.

  ↳ For example, Netflix processing billions of user interactions to suggest specific shows to individual users.

- Descriptive ~~Ann~~ Analytics: the process of using current and historical data to identify trends and relationships. Summarising historical data to describe what has happened.

  ↳ For example, An annual report showing that company revenue grew by 15% last year.

- Diagnostic Analytics: is a branch of business analytics that analyzes historical datasets to uncover root causes, patterns, and relationships. Examining data to understand the root cause of past events, answering the question: why did it happen?

  ↳ For example, a retail company investigating a sudden drop in online sales. They use diagnostic analytics to analyse data points like website traffic, user behaviour, marketing campaign performance, and regional sales data.

- Predictive Analytics: Using historical data and statistical modelling to forecast and identify the likelihood of future outcomes.

↳ For example, GPS apps (like Google maps) analysing traffic and suggesting a new route to save you 10 minutes.

• Data Literacy: the ability to read, work with, analyse, and communicate data in a meaningful way.

↳ For example, an employee being able to explain a chart to their team without misinterpreting the axes.

2. Data, Measurement & Reporting

• Structured Data: is a highly organised data and follows a predetermined schema. For example, an Excel spreadsheet or a SQL database).

• Unstructured Data: information or data that has no predefined format or clear structure. For example, emails, videos, or social media posts.

• Data warehouse: a central storage system used for reporting and data analysis, that collects data from many different sources and keeps it in one place. For example, Retail stores integrate POS data, website clicks, and loyalty programs to analyse buying trends, predict demand spikes, and manage supply chain inventory.

• Data mining: the process of analysing large datasets to discover patterns, trends, and useful information that can help decision making. For example, Fraud detection in banking: identifying unusual patterns in customer transactions such as large withdrawals or purchases in different locations, banks can flag and prevent fraud before it escalates.

• Data visualisation: The graphical representation of information to make data easier to understand. For example, an employee presenting about the companies increase in sales revenues using a chart (or graph).

• Dashboard : a visual tool that tracks and displays key performance indicators (KPIs) and key data points to monitor the health of a business. For example, a sales performance dashboard used to monitor sales activities, revenue tracking, and team analytics to enhance forecasting.

• Metric : A quantitative measure of data. For example, total website visits.

• Key Performance Indicator (KPI) : A critical metric that measures performance against a specific strategic goal. For example, HR uses the employee turnover rate to analyse the percentage of employees leaving the company.

• Benchmark : A standard or point of reference against which things may be compared. For example, Comparing your company's 10% profit margin against the industry average of 8%.

• Segmentation : Dividing a large dataset or customer base into smaller groups based on shared characteristics. For example, a car manufacturer targeting luxury vehicles to individuals with high income, while focusing SUVs on households with children.

3. Customer & Sales KPIs

• Conversion Rate : the percentage of ~~visitors~~ users who complete a desired action (like buying a product) out of the total visitors. For example, for an e-commerce purchase where there is 1000 visitors to a product page and 10 purchases results in a 1% conversion rate.

• Customer Acquisition Cost (CAC) : the total cost of sales and marketing efforts required to gain a new customer.

• Customer Lifetime Value (CLV) : The total revenue a business can

expect from a single customer throughout the entire relationship.

- Churn Rate: the percentage of customers who stop using your product or service during a certain time frame.

- Customer Retention Rate: The percentage of customers who remain with the company over a period.

- Net Promoter Score (NPS): A gold-standard metric for customer loyalty based on one question - how likely are you to recommend us to a friend?

- Average Order Value (AOV): The average rand or dollar amount spent each time a customer places an order.

- Sales Funnel: a visual representation of the customer journey, mapping the stages of a prospect moves through from initial brand awareness to final purchase.

4. Marketing Campaign & Analysis

- Marketing campaign: is a planned, strategic sequence of activities designed to promote a specific product, service, or brand over a set period of time. Such as raising awareness for a new product.

- Post-Campaign Analysis: is the process of reviewing, interpreting, and reporting on data after a marketing campaign concludes.

- Incremental Revenue: is the additional revenue generated specifically because of a particular marketing action or campaign, above the baseline revenue that would have occurred naturally.

• Return on Investment (ROI): is a measure used to evaluate the efficiency or profitability of an investment relative to its cost.

• Return on Ad Spend (ROAS): is a marketing ~~measure~~ metric that measures the amount of revenue generated for every rand / dollar spent on advertising.

• Click-Through Rate (CTR): the percentage of people who click on a link, email, or ad after seeing it.

• Bounce Rate: the percentage of visitors who enter a website and leave without visiting any other pages.

• A/B Testing: a method of comparing two versions of a webpage or ad against each other to see which performs better.

• Lead Generation: the process of attracting potential customers (leads) and gathering their contact information to move them through the sales funnel.

• Marketing Attribution: an analytical process of identifying which marketing touchpoints and channels contribute to a final conversion or purchase.

5. Reference List:

~~• https://~~

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